

Brain Tasks DevOps Project

1. Project Overview

This project demonstrates a complete DevOps pipeline by deploying a React-based application on AWS using Docker, Amazon ECR, AWS CodeBuild, and Amazon EKS.

2. Architecture

GitHub → CodeBuild → Docker → Amazon ECR → Amazon EKS → LoadBalancer

3. Tools Used

GitHub, Docker, AWS CodeBuild, Amazon ECR, Amazon EKS, kubectl, eksctl, CloudWatch

4. Application Setup

Cloned application from GitHub and verified it runs on port 3000 using 'serve'.

5. Dockerization

Created Dockerfile using AWS public ECR nginx image to avoid Docker Hub rate limits.

6. ECR Setup

Created ECR repository and pushed Docker image using AWS CLI.

7. EKS Cluster Setup

Created EKS cluster using eksctl and configured kubectl to interact with cluster.

8. Kubernetes Deployment

Deployed application using deployment.yaml and service.yaml. Service type used: LoadBalancer.

9. CI/CD Pipeline

Configured AWS CodeBuild to automate Docker build and push to ECR.

10. Monitoring

Used AWS CloudWatch logs to monitor build and deployment.

11. Troubleshooting

Resolved issues like GitHub connection errors, IAM permission issues, Docker Hub rate limits, and ECR repository errors.

12. Final Output

Application URL: <http://a70808154209647d79fbbc58b8d6935e-163265015.us-east-1.elb.amazonaws.com/>

🔗 Create instance in Amazon EC2

Recommended:

- OS: Amazon Linux 2
- Instance: t3.medium
- Storage: 20GB

Security Group:

- Port 22 (SSH)
- Port 3000 (App testing)
- Port 80 (optional)

☐ STEP 2: Install All Required Tools

```
sudo yum update -y
```

Git

```
sudo yum install git -y
```

NodeJS

```
curl -fsSL https://rpm.nodesource.com/setup_18.x | sudo bash -
```

```
sudo yum install -y nodejs
```

Docker

```
sudo yum install docker -y
```

```
sudo systemctl start docker
```

```
sudo systemctl enable docker
```

```
sudo usermod -aG docker ec2-user
```

🔗 Logout & login again after this step.

Verify: docker -version

```
[ec2-user@ip-172-31-42-225 ~]$ docker --version
Docker version 25.0.14, build 0bab007
[ec2-user@ip-172-31-42-225 ~]$
```

AWS CLI

sudo yum install aws-cli -y

kubectl

curl -o kubectl https://s3.us-west-2.amazonaws.com/amazon-eks/1.27.0/2023-11-14/bin/linux/amd64/kubectl

chmod +x kubectl

sudo mv kubectl /usr/local/bin/

eksctl

curl --silent --location

"https://github.com/weaveworks/eksctl/releases/latest/download/eksctl_\$(uname -s)_amd64.tar.gz" | tar xz -C /tmp

sudo mv /tmp/eksctl /usr/local/bin

Step 3. Clone & Run Application (Port 3000)

git clone https://github.com/Vennilavanguvi/Brain-Tasks-App.git
cd Brain-Tasks-App

```
[ec2-user@ip-172-31-42-225 ~]$ git clone https://github.com/Vennilavanguvi/Brain-Tasks-App.git
Cloning into 'Brain-Tasks-App'...
remote: Enumerating objects: 11, done.
remote: Counting objects: 100% (1/1), done.
remote: Total 11 (delta 0), reused 0 (delta 0), pack-reused 10 (from 1)
Receiving objects: 100% (11/11), 101.89 KiB | 20.38 MiB/s, done.
[ec2-user@ip-172-31-42-225 ~]$ cd Brain-Tasks-App
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ ls
README.md  dist
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$
```

Install serve

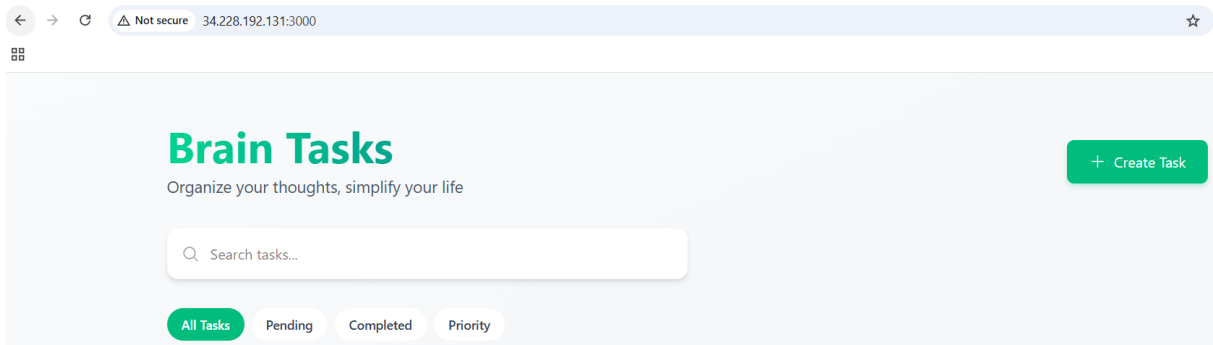
npm install -g serve

Run app

serve -s dist -l 3000

 Verify:

<http://34.228.192.131:3000>



STEP 4: Dockerize App

Create Dockerfile – vi Dockerfile

```
FROM public.ecr.aws/nginx/nginx:alpine
```

```
COPY dist/ /usr/share/nginx/html/
```

```
EXPOSE 80
```

```
CMD ["nginx", "-g", "daemon off;"]
```

Build & Run

```
docker build -t brain-tasks-app .
```

```
docker run -d -p 3000:80 brain-tasks-app
```

STEP 5: Push to ECR

To Create a IAM user

IAM > Users > Create user

Step 1: Specify user details

Specify user details

User details

User name

EKS_user

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)

☐ Provide user access to the AWS Management Console - optional

In addition to console access, users with `SignInLocalDevelopmentAccess` permissions can use the same console credentials for programmatic access without the need for access keys.

Info: If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. [Learn more](#)

Cancel Next

IAM > Users > Create user

Step 2: Set permissions

Set permissions

Permissions options

☐ Add user to group

Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ Copy permissions

Copy all group memberships, attached managed policies, and inline policies from an existing user.

☒ Attach policies directly

Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (1/1480)

Choose one or more policies to attach to your new user.

Filter by Type: All types 29 matches

Search: amazonec

Policy name	Type	Attached entities
☒ AmazonEC2ContainerRegistryFullAccess	AWS managed	0

Create policy

IAM > Users > Create user

Step 3: Review and create

Review and create

Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.

User details

User name: EKS_user

Console password type: None

Require password reset: No

Permissions summary

Name	Type	Used as
AmazonEC2ContainerRegistryFullAccess	AWS managed	Permissions policy

Tags - optional

Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags.

Cancel Previous Create user

```
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ aws configure
AWS Access Key ID [None]: AKIAZUJYQDXNDM4DL3NW
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]: json
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$
```

aws ecr create-repository --repository-name brain-tasks-app

```
aws ecr get-login-password --region us-east-1 | \
```

```
docker login --username AWS --password-stdin 662080331226.dkr.ecr.us-east-1.amazonaws.com
```

```
docker tag brain-tasks-app 662080331226.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:latest
```

```
docker push 662080331226.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:latest
```

```
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin 662080331226.dkr.ecr.us-east-1.amazonaws.com
WARNING! Your password will be stored unencrypted in /home/ec2-user/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ docker images
REPOSITORY          TAG         IMAGE ID      CREATED       SIZE
brain-tasks-app     latest     5ce86268199b  9 minutes ago 62.5MB
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ docker tag brain-tasks-app:latest 662080331226.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:latest
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ docker push 662080331226.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:latest
The push refers to repository [662080331226.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app]
eba317197e43: Pushed
36c803a308d1: Pushed
2b12ccfb247a: Pushed
646d544dc619: Pushed
f7b74afe8798: Pushed
00a43f630638: Pushed
f4eddaa4c303: Pushed
e6082cdc2516: Pushed
29df493baa13: Pushed
latest: digest: sha256:41f43c3e097729c302658e08e661b098c32069a123a69ce37ac13a66b3d35eb3 size: 2199
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$
```

STEP 6: Create EKS Cluster

```
eksctl create cluster \
```

```
--name brain-tasks-cluster-2 \
```

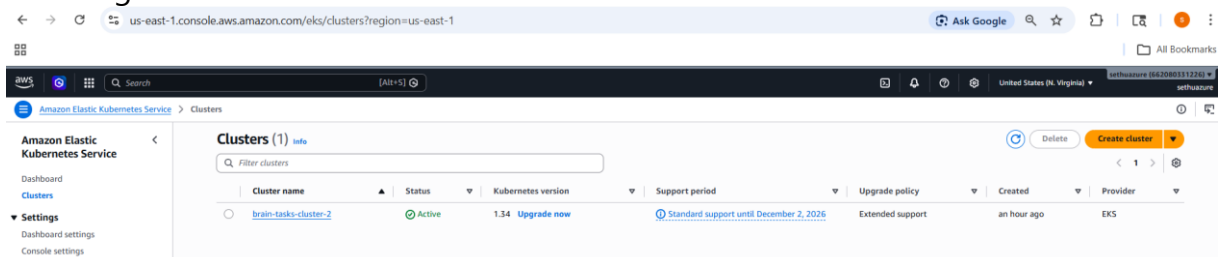
```
--region us-east-1 \
```

```
--nodegroup-name standard-workers \
```

```
--node-type t3.small \
```

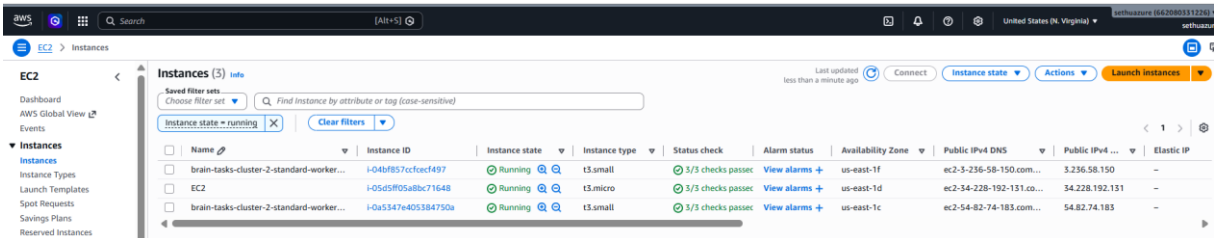
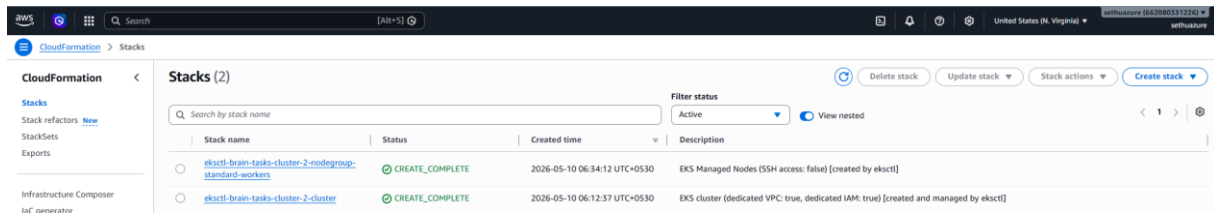
```
--nodes 2 \
```

```
--managed
```



The screenshot shows the AWS Management Console for the Amazon Elastic Kubernetes Service (EKS). The 'Clusters' page is active, displaying a table with one cluster named 'brain-tasks-cluster-2'. The cluster is in the 'us-east-1' region and is currently 'Active'. The table also shows the Kubernetes version (1.34), the support period (Standard support until December 2, 2026), the upgrade policy (Extended support), the creation time (an hour ago), and the provider (EKS). The console also shows the 'Settings' section on the left, including 'Dashboard settings' and 'Console settings'.

Cluster name	Status	Kubernetes version	Support period	Upgrade policy	Created	Provider
brain-tasks-cluster-2	Active	1.34	Standard support until December 2, 2026	Extended support	an hour ago	EKS



Connect kubectl to cluster

aws eks update-kubeconfig \

--region us-east-1 \

--name brain-tasks-cluster-2

Verify nodes

kubectl get nodes

```
[ec2-user@ip-172-31-42-225 ~]$ kubectl get pods -n kube-system
NAME                                READY    STATUS    RESTARTS   AGE
aws-node-cz5q2                      2/2     Running   0           2m32s
aws-node-z89ph                      2/2     Running   0           2m32s
coredns-566b9b9d-gl2g5             1/1     Running   0           2m3s
coredns-566b9b9d-jxnmz             1/1     Running   0           2m3s
kube-proxy-tvf48                   1/1     Running   0          116s
kube-proxy-tzlrz                   1/1     Running   0          116s
[ec2-user@ip-172-31-42-225 ~]$ kubectl get nodes
NAME                                STATUS    ROLES    AGE    VERSION
ip-192-168-26-203.ec2.internal      Ready    <none>   28m    v1.34.7-eks-4136f65
ip-192-168-46-131.ec2.internal      Ready    <none>   28m    v1.34.7-eks-4136f65
[ec2-user@ip-172-31-42-225 ~]$
```

STEP 7: Deploy to Kubernetes

Vi deployment.yaml


```

apiVersion: apps/v1
kind: Deployment
metadata:
  name: brain-tasks-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: brain-tasks
  template:
    metadata:
      labels:
        app: brain-tasks
    spec:
      containers:
        - name: brain-tasks
          image: 662080331226.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:latest
          ports:
            - containerPort: 3000

```

Vi service.yaml

```

apiVersion: v1
kind: Service
metadata:
  name: brain-tasks-service
spec:
  type: LoadBalancer
  selector:
    app: brain-tasks
  ports:
    - port: 80
      targetPort: 3000
~

```

kubectl apply -f deployment.yaml

kubectl apply -f service.yaml

```

[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ vi deployment.yaml
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ vi service.yaml
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ kubectl apply -f deployment.yaml
deployment.apps/brain-tasks-deployment created
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ kubectl apply -f service.yaml
service/brain-tasks-service created
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ kubectl get svc

```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
brain-tasks-service	LoadBalancer	10.100.83.115	ae9156d4db3eb408bb0f64c8ddd2eb73-958218197.us-east-1.elb.amazonaws.com	80:32221/TCP	13s
kubernetes	ClusterIP	10.100.0.1	<none>	443/TCP	52m

🔗 Get URL:

kubectl get svc

ae9156d4db3eb408bb0f64c8ddd2eb73-958218197.us-east-1.elb.amazonaws.com

8. Version Control (GitHub)



STEP 1: Create GitHub Repository

Go to 🔗 GitHub

- Click New Repository
- Fill details:

- **Repository name:** brain-tasks-devops-new
- **Visibility:** Public (recommended for easy AWS access)
- ☒ **Add README** (optional)

▪ Click **Create Repository**

STEP 2: Clone Repo to Your System

git init

git clone https://github.com/sethuraman98/brain-tasks-devops-new.git

cd brain-tasks-devops

```
Last login: Sun May 10 01:24:54 2026 from 27.5.74.173
[ec2-user@ip-172-31-42-225 ~]$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/ec2-user/.git/
[ec2-user@ip-172-31-42-225 ~]$ git clone https://github.com/sethuraman98/brain-tasks-devops-new.git
Cloning into 'brain-tasks-devops-new'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
[ec2-user@ip-172-31-42-225 ~]$ cd Brain-Tasks-App/
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ ls
Dockerfile  README.md  deployment.yaml  dist  service.yaml
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ vi buildspec.yml
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ vi buildspec.yml
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ ls
Dockerfile  README.md  buildspec.yml  deployment.yaml  dist  service.yaml
```

STEP 3: Create Required Files

touch Dockerfile buildspec.yml deployment.yaml service.yaml

We have to create buildspec.yml only remaining file already created and I just moved files from existing directory to new directory

```
version: 0.2
phases:
  install:
    commands:
      - echo Installing dependencies
      - aws --version
  pre_build:
    commands:
      - echo Logging into ECR
      - aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin 662080331226.dkr.ecr.us-east-1.amazonaws.com
  build:
    commands:
      - echo Build started
      - docker build -t brain-tasks .
      - docker tag brain-tasks:latest 662080331226.dkr.ecr.us-east-1.amazonaws.com/brain-tasks:latest
  post_build:
    commands:
      - echo Pushing image
      - docker push 662080331226.dkr.ecr.us-east-1.amazonaws.com/brain-tasks:latest
```

STEP 5: Clone your app inside this repo:

git clone https://github.com/Vennilavanguvi/Brain-Tasks-App.git app

Then move files:

```
mv app/* .
```

```
rm -rf app
```

🚀 STEP 6: Commit & Push

```
git add .
```

```
git commit -m "Final DevOps setup"
```

```
git push origin main
```

```
[ec2-user@ip-172-31-42-225 brain-tasks-devops-new]$ ls
Dockerfile README.md buildspec.yml deployment.yaml dist service.yaml
[ec2-user@ip-172-31-42-225 brain-tasks-devops-new]$ git add .
[ec2-user@ip-172-31-42-225 brain-tasks-devops-new]$ git commit -m "Final DevOps setup"
[main d703ccd] Final DevOps setup
Committer: EC2 Default User <ec2-user@ip-172-31-42-225.ec2.internal>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

5 files changed, 149 insertions(+), 1 deletion(-)
create mode 100644 Dockerfile
create mode 100644 buildspec.yml
create mode 100644 deployment.yaml
create mode 100644 service.yaml
[ec2-user@ip-172-31-42-225 brain-tasks-devops-new]$ git push origin main
Username for 'https://github.com': sethuraman98
Password for 'https://sethuraman98@github.com':
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 2 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 2.00 KiB | 2.00 MiB/s, done.
Total 7 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/sethuraman98/brain-tasks-devops-new.git
e099145..d703ccd main -> main
[ec2-user@ip-172-31-42-225 brain-tasks-devops-new]$ |
```

← → ↺ 🔍 github.com/sethuraman98/brain-tasks-devops-new

☰ sethuraman98 / brain-tasks-devops-new 🔍 Type ↗

🔗 Code 📄 Issues 🔄 Pull requests 🛠 Agents 📁 Actions 📁 Projects 📖 Wiki 🔒 Security and quality 📊 Insights ⚙ Settings

📁 brain-tasks-devops-new Public 📌 Pin 👁 Watch 0

👤 main 1 Branch 0 Tags 🔍 Go to file T Add file <> Code

EC2 Default User	Final DevOps setup	d703ccd · 1 minute ago	3 Commits
dist	Final static deployment setup	7 minutes ago	
Dockerfile	Final DevOps setup	1 minute ago	
README.md	Final DevOps setup	1 minute ago	
buildspec.yml	Final DevOps setup	1 minute ago	
deployment.yaml	Final DevOps setup	1 minute ago	
service.yaml	Final DevOps setup	1 minute ago	

```
git add .
```

```
git commit -m "Initial commit"
```

```
git push -u origin main
```

📦 9. CodePipeline Setup

Pipeline Stages:

1. Source
 - GitHub repo
2. Build
 - AWS CodeBuild project
3. Deploy
 - Use CodeBuild or EKS deploy stage

👉 Ensure IAM roles allow:

- ECR access
- EKS access
- CloudWatch logs

Create CodeBuild Project

Go to:

👉 AWS CodeBuild

Create build project

Project Name : brain-tasks-build

Source Provider: GitHub

Repository: brain-tasks-devops-new

Branch: main

Environment Image: Managed image

Operating System: Amazon Linux

Runtime: Standard

Image: Latest standard image.

VERY IMPORTANT

Enable:

☒ Privileged

Without this:

✗ Docker build fails

Service Role: New service role

Role name: codebuild-brain-tasks-role

Buildspec: buildspec.yml

Artifacts: No artifacts

Logs: ☒ CloudWatch Logs

The screenshot shows the 'Create build project' page in the AWS CodeBuild console. The 'Project configuration' section has 'Project name' set to 'brain-tasks-deploy'. Under 'Project type', 'Default project' is selected. The 'Additional configuration' section is collapsed. The 'Source' section is expanded, showing 'Source 1 - Primary' with 'Source provider' set to 'GitHub'. The 'Credential' section shows a success message: 'Your account is successfully connected by using an AWS managed GitHub App. Manage account credentials.' The 'Repository' section has 'Repository in my GitHub account' selected, and the repository path is 'https://github.com/sethraman98/brain-tasks-devops-new'.

The screenshot shows the 'Environment' section of the 'Create build project' page. Under 'Provisioning model', 'On-demand' is selected. Under 'Environment image', 'Managed image' is selected. Under 'Compute', 'EC2' is selected. Under 'Running mode', 'Container' is selected. The 'Operating system' is 'Amazon Linux', 'Runtime(s)' is 'Standard', and 'Image' is 'aws/codebuild/amazonlinux-x86_64-standard-5.0'.

The screenshot shows the 'brain-tasks-build' project details page in the AWS CodeBuild console. A green success banner at the top states: 'Success Project details for build project brain-tasks-build successfully updated.' The 'Configuration' section shows 'Source provider' as 'GitHub', 'Primary repository' as 'sethraman98/brain-tasks-devops-new', 'Artifacts upload location' as '-', and 'Service role' as 'arn:aws:iam::662080311226:role/service-role/codebuild-brain-tasks-role'. The 'Public builds' are 'Disabled'. The left sidebar shows the navigation menu with 'Build project' highlighted. The bottom tabs include 'Build history', 'Batch history', 'Project details' (active), 'Build triggers', 'Metrics', and 'Debug sessions'.

Attach IAM Policies

Go to:

👉 AWS Identity and Access Management → Roles

Search:

codebuild-brain-tasks-role

Attach:

AmazonEC2ContainerRegistryFullAccess

CloudWatchLogsFullAccess

AdministratorAccess

Identity and Access Management (IAM)

codebuild-brain-tasks-role

Policies have been successfully attached to role.

codebuild-brain-tasks-role

Summary

Creation date
May 18, 2026, 09:35 (UTC+05:30)

ARN
arn:aws:iam:662080351226:role/service-role/codebuild-brain-tasks-role

Last activity
-

Maximum session duration
1 hour

Permissions | Trust relationships | Tags | Last Accessed | Revoke sessions

Permissions policies (4)

You can attach up to 10 managed policies.

Policy name	Type	Attached entities
AdministratorAccess	AWS managed - job function	3
AmazonEC2ContainerRegistryFullAccess	AWS managed	3
CloudWatchLogsFullAccess	AWS managed	2
CodeBuildRolePolicy-codebuild-brain-tasks-role-us-east-1	Customer managed	1

Start Build

Cloudwatch

CloudWatch Monitoring

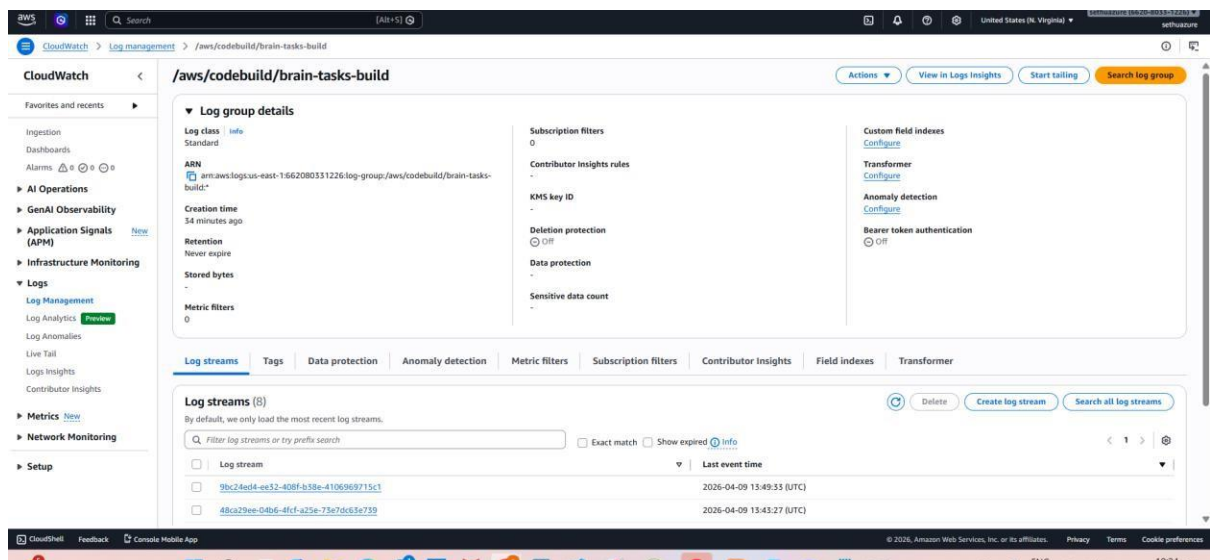
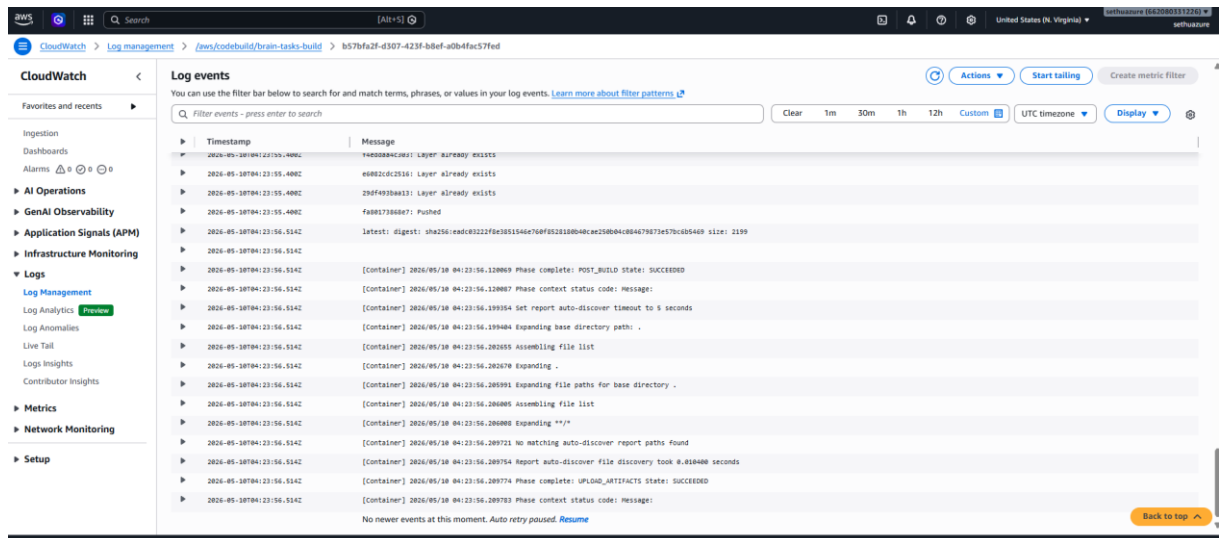
CloudWatch Logs were enabled for AWS CodeBuild to monitor build execution, Docker image push activity, and deployment-related logs. The logs confirm successful completion of the build and post-build stages.

Build logs

🟢 Succeeded

Start time: 32 minutes ago

Current phase: COMPLETED



“I implemented a full DevOps CI/CD pipeline using AWS services including Docker, ECR, EKS, and CodeBuild to deploy a React application, which is accessible via a LoadBalancer URL.”

Troubleshooting:

I didn't get the output. I have to check the ports.

← ↻ ae9156d4db3eb408bb0f64c8ddd2eb73-958218197.us-east-1.elb.amazonaws.com



This page isn't working right now

ae9156d4db3eb408bb0f64c8ddd2eb73-958218197.us-east-1.elb.amazonaws.com didn't send any data.

ERR_EMPTY_RESPONSE

Refresh

Updated deployment.yaml file

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: brain-tasks-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: brain-tasks
  template:
    metadata:
      labels:
        app: brain-tasks
    spec:
      containers:
        - name: brain-tasks
          image: 662080331226.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:latest
          ports:
            - containerPort: 80
```

Updated service.yaml file

```
apiVersion: v1
kind: Service
metadata:
  name: brain-tasks-service
spec:
  type: LoadBalancer
  selector:
    app: brain-tasks
  ports:
    - port: 80
      targetPort: 80
```

~

Run:

```
kubectl apply -f deployment.yaml
```

```
kubectl apply -f service.yaml
```

Restart Deployment

```
kubectl rollout restart deployment brain-tasks-deployment
```

Verify Pods

```
kubectl get pods
```

```
kubectl get svc
```

```
[ec2-user@ip-172-31-42-225 brain-tasks-devops-new]$ vi deployment.yaml
[ec2-user@ip-172-31-42-225 brain-tasks-devops-new]$ vi service.yaml
[ec2-user@ip-172-31-42-225 brain-tasks-devops-new]$ kubectl apply -f deployment.yaml
kubectl apply -f service.yaml
deployment.apps/brain-tasks-deployment configured
service/brain-tasks-service configured
[ec2-user@ip-172-31-42-225 brain-tasks-devops-new]$ kubectl rollout restart deployment brain-tasks-deployment
deployment.apps/brain-tasks-deployment restarted
[ec2-user@ip-172-31-42-225 brain-tasks-devops-new]$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
brain-tasks-deployment-7746788b9-mbzqh  1/1     Running   0           8s
brain-tasks-deployment-7746788b9-p48j5  1/1     Running   0           7s
[ec2-user@ip-172-31-42-225 brain-tasks-devops-new]$ kubectl get svc
NAME                                TYPE               CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
brain-tasks-service                LoadBalancer      10.100.83.115   ae9156d4db3eb408bb0f64c8ddd2eb73-958218197.us-east-1.elb.amazonaws.com  80:32221/TCP    3h28m
kubernetes                         ClusterIP          10.100.0.1      <none>           443/TCP          4h13m
[ec2-user@ip-172-31-42-225 brain-tasks-devops-new]$ |
```

ApplicationLoadBalancerURL:

<http://ae9156d4db3eb408bb0f64c8ddd2eb73-958218197.us-east-1.elb.amazonaws.com>

Final Output:

